Philosophical Problems

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Introduction

Philosophical ideas can, as the history of philosophy shows, have a twofold significance. They can be attempts at the formulation, the treatment, or the solution of certain problems, and they can stand as symptoms of certain tendencies in the intellectual life of man. There is a constant interaction between these two sides of philosophy, in that it may stand as a symptom of a distinctive intellectual movement that a problem is formulated and treated in a particular way, while on the other hand the sting of the problems can release intellectual movements that would otherwise never come to be. There is expressed in this interaction an inner connection between personal life and scientific inquiry, one which asserts itself more or less in every scientific field, but which in the nature of the case will come forward quite particularly as we approach the border regions of the sciences. It comes forward more in the human sciences than in natural science, and it comes forward most of all in philosophy, whose problems are essentially border problems.

It is admittedly no uncommon assumption that there is a conflict between personality and scientific inquiry — that one had best divest oneself of one's personality when one wishes to think scientifically, and that on the other hand scientific methods and results signify nothing for personal life in its free development. I myself in my youth paid homage to such an assumption, under the overwhelming influence of S. Kierkegaard's ideas. That I have come more and more away from it is, I venture to say, because I have learnt to know both science and personal life better.

The need for scientific inquiry is a special form of the need for accord with oneself under all the manifold and shifting experiences. It expresses itself both in the formal sciences and in the real — both as a need to form series of thoughts in which the one member develops with inner necessity out of the other, and as a need to connect the experiences before us in such a way that as comprehensive and as firm a continuity as possible is attained. Personality consists first and foremost in an inner unity and connection among all representations, feelings, and strivings. It is then no sacrifice that personality makes when it stakes its life upon inquiry. It brings forth in science

— as in art — an objective counter-image of itself. Only thus is the inward character that the joy of knowledge can take on to be understood, the *amor intellectualis* that the inquirer feels in working his way through to a standpoint from which the particular can be seen as a member of a great connection. The sacred fire which ever anew sets thought in motion, in spite of all that can damp and hinder it, is understood only when one has this in view. And personality on its side needs to be purified by bowing before the objective connection of thought, the fixed order of things, within which every single being has its determinate place. Freedom is won through truth. It belongs to the nobility of personality that it can by its own power find the great laws which condition its subsistence, determine the wishes that arise in it, and determine the conditions under which those wishes can be realized. There is a science of personal life itself, as there is of everything else; this science is perhaps more imperfect than other science, but it is presupposed in every serious discussion of the forms and demands of personal life. Even Charles Renouvier, the philosopher now living who lays the greatest weight on the opposition between necessity of thought and personality, nevertheless requires "a rational concept of personality"[1] and thus cannot reject every analogy between them.

Such an analogy will come clearly forward when one attempts to formulate the principal problems with which philosophical inquiry occupies itself. These problems are posed at one and the same time from the side of personality and from the side of science.

By a purely historical route I have sought, in my work *Den nyere Filosofis Historie*, to show that there are four such principal problems, namely I) the problem of the nature of conscious life (the psychological problem), II) the problem of the validity of knowledge (the logical problem), III) the problem of the nature of existence (the cosmological problem), and IV) the problem of valuation (the ethical-religious problem).[2] It will now be my task in this treatise to show the inner connection among these problems. It is one and the same fundamental problem that appears in them in different forms and applications.

It is very different motives that can lead to philosophical inquiry. — Very often it is an ethical-religious, and so a practically personal, interest that leads to philosophy. One then begins with the fourth problem. But it soon shows itself that the treatment of that problem requires insight into the nature of conscious life, into the conditions of knowledge, and into the constitution of the existence in which the valuing personality

[1] Ch. Renouvier: *Les dilemmes de la métaphysique pure.* Paris 1901, p. 202.

[2] These four problems contain the ancient tripartition into Logic, Physics [i.e. Cosmology] and Ethics, deriving from Plato's school (according to *Sextus Empiricus*: *Adv. mathematicos* VII, 16, from Xenocrates), with the addition of the psychological problem, which in more recent times has pushed itself forward as an independent point of departure.

exists. — If it is an interest in observation that lies at the ground, one will begin with the psychological problem, as has happened in empirical philosophy in its various forms. — If on the other hand it is a need to distinguish between what can be known and what cannot be known, one will begin with the problem of knowledge; this is the road critical philosophy takes. — If one has full confidence in the power of thought and seeks a completed world-view, one will — like the dogmatic and speculative tendencies — begin with the problem of existence. — It will of course not be a matter of indifference, for the kind and manner of the treatment of the problems, which problem one begins with; the one problem will easily be put in the shade by the other. For a comparative doctrine of problems, such as I here seek to contribute to, what matters is to let each single problem come into its full right. I proceed, then, in this way: in keeping with the analogy indicated above between personality and science, I first bring forward the problem of the nature of personal life, that is, the psychological problem, and then pass over to investigating the nature of scientific knowledge, that is, the problem of knowledge. When the problem of existence is placed third in the series, the transition to it is natural both from personality, which stands as a single part of the whole of existence, and from science, whose task it is to lead to a world-view. The three problems named so far would be posed even if man were a purely intellectual, merely knowing being. The fourth problem is posed on account of the relation in which man, as a feeling and willing being, stands to existence. Thus the principal problems of thought hang together with man's theoretical and practical interests.

But more important than the question of the order in which the problems are presented is the question whether they can be led back to one pervading fundamental problem. The possibility of this I find in the significance which the relation between continuity and discontinuity has within each of the several problems. This relation expresses the deepest interest both of personality and of science. In both domains there is a striving, as already remarked, after unity and connection, and the discontinuous stands so far as an obstacle to be overcome. But on the other hand it is discontinuity (difference in time, in degree and in place, difference of quality, difference of individuality) that everywhere, in the domain of science as in that of life, brings new content, releases the bound forces, and sets the great tasks. Neither of the two elements stands, then, as from the outset the only one justified. It will have its unquestionable interest to follow the relation between them under the four different points of view that our four principal problems indicate.

In the philosophy of the nineteenth century the significance of the problem of continuity comes forward in a characteristic way, in that the different tendencies plead the cause of continuity by turns. In the first half of the century philosophical idealism asserted in its fashion the continuity in existence and looked down on the science of experience on account of the latter's fragmentary character, while positivism (as Comte and Stuart Mill maintained it) impressed the discontinuity between the different groups of phenomena. At the century's close it is realism that, with the help of the hypothesis of evolution, would assert continuity, while the idealist tendency is inclined to lay weight on the unavoidable discontinuity in our knowledge.[3] Thus the tendencies change places in the great single combat in which the fight for truth is to be waged. When a significant thought can no longer be carried through from one point of view, inquiry involuntarily seeks a new point of view, and thought gains in clarity through these shifting points of view. The different tendencies relieve one another, like the runners at the ancient torch festival, but the torch is the same. And if none of the tendencies that have so far come forward should have done full justice to the fundamental thought in all the philosophical problems, there is so much the more reason to work at forming it with greater clarity.

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[3]Compare, besides the work of Renouvier's named above (Note 1), the interesting writings of *Emile Boutroux*: *De la contingence des lois de la nature.* 2. éd. Paris 1895, and *De l'idée de loi naturelle dans la science et la philosophie contemporaines.* Paris 1895. — Related is the standpoint of *Rudolph Eucken* (*Der Kampf um einen geistigen Lebensinhalt.* Leipzig 1896).

I. The Problem of Consciousness

1. When we begin the discussion of the philosophical problems by investigating the concept of personality itself, we begin with psychology, and it is then first of all a matter of determining psychology's relation to philosophy. We can here no doubt leave aside the view, which Herbart and Lotze still maintained, that psychology should be dependent on the view one arrives at concerning the problem of existence, and so dependent on "metaphysics" or cosmology. On all sides an effort is at work to assert the independence of psychology. But even where this effort prevails, there have still been extremely different views about psychology's position.

There are those who hold that psychology is one with philosophy as a whole, and that epistemology, metaphysics, aesthetics and ethics are its different parts. Such a standpoint was taken in earlier times by Fries and Beneke, and is taken in our own day by Th. Lipps.[4] This standpoint is untenable, partly because personal life is only one of the subjects that can be an object of philosophizing, partly because psychology itself, like all other science, presupposes the general forms and principles of knowledge. The problem of consciousness thus necessarily points beyond itself to the problem of existence and the problem of knowledge, and for all its independence psychology is only one discipline among several. Psychology can then only be a part of philosophy, not constitute the whole of philosophy. It is legitimate to begin with psychology, since psychology describes the place, and the presuppositions, from which we seek to orient ourselves in existence.

But does psychology belong to philosophy at all? Is it not a special science which in and for itself has no more to do with philosophy than natural science or history has? The special roads along which psychological knowledge is sought in the most recent period — the experimental, the physiological and the historical method — do after all seem to testify that psychology is now in the course of becoming a special science and must be separated off from philosophy! — To this the answer is, so far as I can see, that psychology,

[4] Th. Lipps: *Psychologie, Wissenschaft und Leben.* München 1901. On Fries and Beneke see my work *Den nyere Filosofis Historie* II, p. 221 f.; 236; 240.

for all its special methods, always presupposes the capacity for self-observation. It is subjective, descriptive and analysing psychology that sets the tasks which can then be treated in detail with the help of the special methods. There is here a foundation common to psychology as a special science and to philosophy. Philosophy must build on a general concept of the nature of conscious life if it is to be able to treat the problems of knowledge, of existence and of valuation. It is personality's position as knowing and valuing, and as a part of the whole of existence, that these problems concern, and a concept of personality is presupposed which none of those special methods is able to furnish. They investigate single expressions of conscious life, but their investigations must be worked together if a concept of personality is to be formed.

But here precisely the fundamental problem arises in the psychological domain: is it possible by the road of experience to arrive at a concept of personality? Does our conscious life form a whole, a continuum, a little world of its own, or should it be merely an aggregate, a sum of elements and fragments? This is the question that philosophical psychology (or, if one prefers, the philosophy of psychology) raises. And in its treatment of it it relies partly on self-observation and descriptive psychology, partly on the results of the experimental, the physiological and the historical method.

Experience shows us greater and lesser interruptions of our conscious life; there are unconscious intervals between the conscious states. And when we investigate the single state of consciousness more closely, we can distinguish within it different elements which can be found again in other states, so that the single state, no less than consciousness as a whole, might seem to be a resultant of a manifold of elements or fragments which stand as what is properly real. Consciousness would then be an aggregate, or a sum of single flashes. A psychical continuum, such as is presupposed in the concept of personality, there would not be.

To this it must first be remarked that it will never be possible to decide with full certainty whether analysis has reached down to the last elements. The possibility always stands open that the elements at which we stop, and out of which we are inclined to think consciousness composed, are themselves composite, and that if we could reach down to still simpler elements, elements of the second or third order, a continuity would appear which could not be exhibited so long as we stopped at the elements of the first order. Our sensations, which one has been inclined to regard as a kind of psychical atoms, nevertheless bear — as recent investigations along various roads have shown — traces of processes of composition of a still more elementary kind than those which observation shows us more directly.[5]

This, however, is only a preliminary consideration, and leads to no settled result, since the defender of discontinuity may very well use it to show that all the continuity that can be exhibited is only preliminary. What is on the other hand of decisive importance is that the so-called psychical elements are always determined by the connection in which they occur, and that it is only by abstraction that one ascribes to them, outside this connection, properties which they have only within the connection itself. It is on this fundamental thought that my presentation of psychology is built. All mental life works in the manner of genius: it brings forth immediately and involuntarily connected phenomena and processes which analysis afterwards seeks, in a more or less artificial way, to dissolve into "elements". As regards sensation we can here appeal to Fechner's law and to the law of contrast, according to which the degree of intensity and the quality of every single sensation are determined by the whole connection in which it comes to be. The connection cannot, then, be conceived as a product of the psychical elements, since these come to be as *these* determinate sensations only in virtue of the connection. Did they occur in another connection, they would not be the same sensations. The relation is analogous as regards representations. The association of representations finds its ultimate explanation in the unnaturalness of isolating the single representations; there is always at work — and to a higher degree, the more vigorous consciousness is — a tendency to supplementation or extension, whereby the single representation enters into combination with other representations according to determinate laws. The so-called "Association-Psychology" (among whose adherents I have myself sometimes wrongly been reckoned) conceived the single representations as independent atoms which were brought into connection with one another in a purely external, mechanical way. But it is precisely the reverse: it is the connection of the whole that in the process of association exercises its power over the single representations, which moreover never occur in a wholly isolated condition, such that their combination could stand as a mechanical product. Analysis here too always presupposes synthesis. In thinking proper this relation shows itself plainly when we compare the formation of judgments and the formation of concepts. These mutually presuppose one another, since our judgments, which are after all combinations of concepts, can only be complete when the concepts combined in them are complete, while conversely the formation of concepts presupposes a series of judgments by which different elements are determined in their mutual relation. It shows itself here again that the whole and the parts mutually determine one another. There are

[5] See my *Psykologi* V A, 2.

no wholly isolated concepts which could only afterwards be combined into judgments.

There expresses itself here an antinomy which hangs closely together with the essence of consciousness and is peculiar to the concept of personality. Consciousness and personality cannot be explained as products of antecedently given elements, any more than organic life can be explained as a product of inorganic elements. But on the other hand the essence of consciousness and of personality, like the essence of organic life, displays itself in a constant gathering-together of *given* elements, elements which they do not themselves originally bring forth. It is this antinomy that makes the coming-into-being of life and of personality such great riddles.[6]

In the foregoing, account has been taken only of the more formal connection within conscious life. But in every consciousness there is a goal towards which there is striving, a guiding interest which takes, or strives to take, everything into its service. This guiding interest — the fundamental value, one might call it — can change in the different ages of life; but there will always anew be a tendency to the development of such a one, and in higher or lower degree it sets its stamp on all the elements of consciousness, gives them their direction. It constitutes the soul proper (when by "spirit" we think chiefly of the formal connection).

It is the relations here brought forward that make the concept of personality have always to stand as the central thought in psychology. When analysis and the special methods make their sections and seek to isolate single elements or moments, that is legitimate, just as it is legitimate to attempt to determine an irrational number by adding decimal to decimal, as far as one can. But the irrational relation between the whole and the elements continues to subsist.

From the idealist side one has often been inclined to take the concept of personality for something finished and to operate with it in cosmological speculations. What is thereby overlooked — and it is what the positivist tendency particularly emphasizes — is that we are constantly at work forming the concept of personality, spelling our way forward to a psychological conception of the whole, just as biology is still constantly spelling out a definition of life. And just as biology, for all its recognition of the peculiarity of the living organism, knows no other method than that which through observation, experiment and analysis seeks to understand the composite processes by the help of the simpler, so psychology, however firmly it holds to the synthetic character of conscious life, can apply no other methods than those which all science applies: observation, experiment and analysis. The concept of personality stands as the ideal towards which the course is

[6] Compare my treatise *Det psykologiske Grundlag for logiske Domme* (Vid. Selsk. Skr. 6. Række), p. 16 f.; 59, and my treatise *Om Vitalisme* (reprinted in "Mindre Arbejder").

set, the standing problem which all the special methods serve to illuminate.[7]

The irrational is here, as everywhere, not merely a barrier but also a task, a task that is set ever anew. Descriptive psychology, which leads one particularly to lay weight on the connection of the whole in and with which mental phenomena appear, will here — especially as regards the higher or more developed phenomena of consciousness — always preserve its independence and its importance in relation to experimental psychology; the latter in the end always receives its tasks from the former. And descriptive psychology serves at the same time as a corrective to experimental psychology, which by its method easily comes to isolate single elements too much, to overlook what is purely involuntary in conscious life, and to lay too great weight on the outward symptoms of states.[8] On the other hand descriptive psychology will be able to draw advantage from experimental psychology, in that what the latter is able to exhibit concerning the more elementary psychological processes can by the road of analogy throw light on the nature of the higher processes and give description greater fullness and accuracy.

Descriptive psychology approaches art and becomes in the end an art; experimental psychology approaches natural science. But there is nevertheless no difference of principle between them. There is not, as MÜNSTERBERG in the most recent period especially has wished to maintain,[9] a gulf between them. The former is to be a "science of value", the latter a natural science; the former is to operate with the "concept of freedom", the latter with the concept of cause. By freedom Münsterberg understands the capacity to act according to purposes. But should then the psychical process by which a human being comes to set himself a purpose and to work at attaining it not be an object for scientific psychology? And can it be understood otherwise than by being investigated in connection with the simpler processes that take place when pleasure and pain, joy and sorrow arise, or when representations are combined and separated? Here there is a mass of experience to be gathered and collected; what matters is to follow a development step by step, to establish the different degrees of consciousness and the differing impulses and motives that co-operate. There will further be a distinction to be drawn between the different individual types that can appear with respect to the kind and direction of the life of will, and a comparative individual psychology will be able to serve as a

[7] Of interest for the standpoint of modern positivism on this problem, in contrast to the standpoint Comte and Stuart Mill took, is ROBERTO ARDIGÒ's work *L'unità della coscienza.* Padova 1898.

[8] Compare my *Psykologi* I, 8 d.

[9] MÜNSTERBERG: *Psychology and Life.* Westminster 1899. — Münsterberg allies himself with the view that Windelband and Rickert have urged, according to which there is a gulf between natural science (to which psychology too is reckoned) and the science of culture. On this view see the apt remarks in WUNDT: *Einleitung in die Philosophie.* Leipzig 1901, p. 65–74, and in GUIDO VILLA: *Psychology and History.* (The Monist, January 1902.)

necessary aid to understanding the development within the consciousness of the single person. It is then a wholly unnatural boundary that is drawn when one lays down a difference of principle between "personal" and "psychophysical" categories — and especially when this opposition is made the ground of an opposition between "the truth of life" and "the truth of science".[10] It is after all the task of science to understand life, even if complete understanding is always an ideal, and it is life that yields science the material of experience on which it builds. Here there is a constant relation of interaction, but no absolute opposition. And there are many intermediate links and intermediate forms between science's separating out of the single elements of existence and life's great interplay of all elements, which in the end only art can present in its whole fullness.

2. Nevertheless great difficulties always remain for the scientific treatment of the phenomena of consciousness. To a real understanding it belongs that a continuity be exhibited in the processes one investigates. It would be psychology's ideal not only that we could win so complete a description of the states of consciousness that each single one of them came to stand as a distinctive member of the whole psychical process, but also that we could reduce the differences between the changing states to forms so simple that the following appeared as continuations, or as transformations, of the preceding. But the prospect of reaching this ideal is barred by the discontinuity which experience seems to establish. There are, after all, unconscious intervals between our conscious states — in fainting, in dreamless sleep (so far as there is such a thing); and there are qualitative differences between the different states and elements of consciousness, so that every single state and every single element seems to come to be out of nothing, if we keep to the strictly psychological consideration. Between the different individual consciousnesses there is in any case a great and conspicuous discontinuity. The one consciousness can be derived from the other still less than the one state of consciousness can be derived from the other.

These psychical relations of discontinuity come forward especially sharply when they are set over against the continuity, and the relations of equivalence, which the material phenomena present and of which men became aware very early. Among the Greek natural philosophers the persistence of matter through all changes was a common assumption, and this assumption has received its confirmation in the chemistry of more recent times. In sharp contrast to it stood the distance between souls among themselves.

[10] Münsterberg, op. cit. p. 282: "If we force the system of science upon the real life, claiming that our life is really a psychophysical phenomenon, we are under the illusion of psychologism. If, on the other hand, we force the views of the real life, the personal categories, upon the scientific psychophysical phenomena, we are under the illusion of mysticism. The result on both are the same. We lose the truth of life and the truth of science."

When Descartes applied the concept of substance both in the psychical and in the physical domain, it was with the great difference that he assumed many soul-substances but only one single material substance.[11] Thereby the psychical discontinuity was pronounced sharply and definitely, in contrast to the physical continuity. It is only in a half or wholly mystical fashion that we find, in the history of philosophy, the continuity of mental life asserted in such a way that the single consciousnesses are thought to stand in the same relation to a universal soul-substance in which the single bodies stand to the universal material substance. So with Averroes, Spinoza and Hegel.[12] In the most recent period it is now not so much the relation of discontinuity between individual consciousnesses as the relation of discontinuity within the single consciousness that draws attention to itself. And there are in particular two considerations which have here led to the problem's being posed more sharply than before.

Physiology has come to full consciousness of its method and of its independence as a science, especially after the reform it underwent about the middle of the nineteenth century. It now consistently requires that every state of the brain be explained from the preceding states of the brain and of the organism — that physiological phenomena in general be explained by physiological causes, in so far as they are not effects from the physical world outside, continued into the organism. On strict natural-scientific method every material state within and outside the organism must find its explanation by being seen as having arisen through the conversion, into a new form, of the energy which expressed itself in the preceding states. When two brain-states succeed one another, the explanation will accordingly first have been reached when it is shown that all the energy which expressed itself in the first state has been converted without remainder into nutritive activity, heat, electricity, motor impulses and so on. It is indeed difficult to

[11]Descartes (*Synopsis meditationum*) teaches that all substances are imperishable, but that it is only *corpus in genere sumptum* that is substance, not for example *corpus humanum*; every soul, by contrast, is *pura substantia*, which does not change, however much its single *accidentia* (thoughts, feelings and expressions of will) may vary. — In his work *Cogitata metaphysica* (II c. 11) Spinoza says (from the Cartesian standpoint which in that work he essentially adopts): *Si ad totam Naturam materiæ attendamus, illi nihil novi accedit; at respectu rerum particularium aliqvo modo potest dici, illi aliquid novi accedere. Qvod an etiam locum habet in rebus spiritualibus, non videtur: nam illa* [i.e. *spiritualia*] *ab invicem ita dependere non apparet* [souls do not depend upon one another as bodies depend upon one another]. Later Spinoza came to another view.

[12]Relying on his interpretation of Aristotle's "active reason", Averroes (on him see Renan: *Averroës et l'Averroïsme.* Paris 1852) teaches that while the single souls arise and perish, there subsists *intellectus universalis*, the world-thought that is at work in the thinking of the single souls. In the Middle Ages this doctrine was renewed, among others, by Siger of Brabant in his *Qvæstiones de anima intellectiva*, recently edited by Mandonnet. (Mandonnet: *Siger de Brabant et l'Averroïsme latin au 13^e^ siècle.* Fribourg 1899.) The whole doctrine of the actual subsistence of spirit goes back to Neoplatonism. (See Plotinos: *Ennead.* V, 9, 6.) — On Spinoza's doctrine of *intellectus infinitus* and *idea dei* see *Den nyere Filosofis Historie* I, p. 290; on Hegel ib. II, p. 165.

imagine such an experiment carried out; but it would mark the carrying through of the physiological method. Neo-vitalism itself knows no other method for the investigation of physiological states. Thus is confirmed Spinoza's saying, that when one appeals to the intervention of the soul in explanation of a bodily state, it is the same as saying that one does not know what the cause is. Even if an experiment should show that at the arising of an organic state energy arose or disappeared without physical equivalent, one would surely sooner believe that there was a fault in the experiment than acquiesce in the result. "The principle of the conservation of energy has", says Maxwell,[13] "acquired so great a scientific weight that no physiologist would have any confidence in an experiment which showed a considerable difference between the work performed by a living being and the reckoning of energy received and expended." — By carrying through the physiological method, understood in this spirit, an ever greater continuity will be exhibited in the organic processes to which the phenomena of consciousness in fact show themselves to be attached. Physiology is therefore far more happily placed with respect to the principle of continuity than psychology will ever be able to be, and it might therefore seem a well-founded demand that Carl Lange put forward when he declared that all psychological definitions ought to be replaced by physiological ones.[14] The psychical phenomena then come to stand as mere provisional symptoms.

The second consideration attaches to the one just adduced, but is of a general epistemological kind. It rests on the proposition that a complete understanding is won only where the relation between cause and effect can be made a relation of identity or of continuity, such that a quantitative equation becomes possible between the phenomena that are connected in the causal relation. It is the ideal concept of cause that is here maintained, in contrast to the empirical concept of cause, by which phenomena that are qualitatively different stand as connected in unavoidable succession. The closer investigation of these two concepts of cause I postpone to the problem of knowledge. Here it need only be remarked that there is no doubt that where the ideal concept of cause can be carried through we obtain a more complete understanding than where we must stop at the empirical concept of cause — and have, perhaps, difficulty enough even

[13] Maxwell: *Scientific papers.* Vol. II. Cambridge 1890, p. 759. Shortly before the passage quoted, Maxwell remarks that whereas the thought that the soul might be found somewhere in the brain by anatomical means has nowadays as a rule been given up, the thought has held on longer that by pursuing the material processes far enough back one will come to a material process brought about by the soul. It is against this possibility that the quoted passage is directed. Compare Spinoza: *Ethica* III, 2, Schol. — His conception of the principle of inertia in his work *Matter and Motion* § 41 leads to a result similar to Maxwell's conception of the principle of energy. — Compare my *Psykologi*[4] p. 30 and my *Psykologiske Undersøgelser*, p. 82–86.

[14] Carl Lange: *Nydelsernes Fysiologi.* København 1899, p. 45.

in applying that. Now both the material and the psychical phenomena present qualitative differences; but in the material domain it is possible to conceive these qualitative differences as differences of quantity, namely by virtue of the law of the conservation of energy and of matter, while nothing corresponding is possible in the psychical domain. The right psychological method is therefore taken to consist in substituting the physiological phenomena for the corresponding psychical ones; that is the only way in which we can put quantitative determinations in the place of qualitative ones and thereby carry through a stringent causal connection. This conception of the psychological problem has been carried through most strictly by Richard Avenarius in his acute work *Kritik der reinen Erfahrung*. He requires there, as the condition of full scientific understanding, a reduction of "the dependent vital series", by which are meant the psychical states, to "the independent vital series", the corresponding physiological states, whereby alone it becomes possible to reduce the empirical differences to the least possible, or, as Avenarius expresses it, to reach "a heterotic minimum". In a similar direction goes Münsterberg's view, as he has expressed it in his work *Psychology and Life*. His principal contention is that since psychical phenomena are not quantitative, they cannot be members of a causal series.[15]

As against the views which thus require a reduction of psychology to physiology in order that a scientific psychology should become possible — which really, then, would abolish psychology in order to make it a science — it must be acknowledged that they can appeal to the factual discontinuity and the qualitative difference between psychical phenomena, which will always more or less set the limit to a strict carrying through of psychology as a science. Whether the road they point out leads to the goal, and whether the method they commend does not contain a palpable self-contradiction, is another matter.

When it is required that the psychological definitions be *replaced* by physiological ones, it is clear that the presupposition of this is that the psychological definitions must first be finished. But to procure those definitions must be psychology's business, and if it cannot procure definite and secure definitions, physiology cannot know what it is that it is to seek an explanation of in the brain: if what is to be replaced is vague and wavering, the replacement will be so accordingly. There will be no security that one has really found the brain-states which correspond to the psychical phenomena one has in view. Psychology's independence must therefore in any case be acknowledged, since it is psychology that here — as a kind of symptomatology — is to set physiology

[15] Münsterberg: *Psychology and Life*. p. 127: "Mental facts, as they are not quantitative, cannot enter into any causal equation."

its tasks. Now for every science of experience it is a long and difficult labour to find conclusive definitions; such are possible only when the science of experience is in essentials completed; they come at the end, not at the beginning, of the investigation. Far too often one operates with unfinished psychological definitions which are regarded as trustworthy points of departure for the physiology of the brain. Thus DESCARTES, and in more recent times LOTZE, regarded the very concept "soul" as so distinct and clear that one could confidently invite physiologists and anatomists to search for "the seat of the soul". In the most recent period FLECHSIG[16] has regarded the concept "association" as so simple and clear, and withal as so independent, that he has been able to think he could find a peculiar place in the brain for the functions that concept designates. That it is nevertheless a highly imperfect concept of association that Flechsig here takes as his ground is seen from the fact that psychological experience never shows us absolutely single psychical elements which are only afterwards to be brought into connection with one another by a peculiar process. One cannot set the process of association in definite contrast to the processes by which the single psychical elements (sensations and representations) come to be. It is a psychological abstraction, not a real psychological definition, that Flechsig operates with. Anatomical objections have been raised from several sides against Flechsig's doctrine of peculiar association-centres, which I cannot judge; but the psychologically unfinished character of his concept of association shows sufficiently how difficult a task it is to "replace" psychological definitions by physiological ones.[17] A principal point for the criticism of Flechsig's theory seems to me to be the following circumstance. When there are psychical elements which do not enter into such an association as might under usual conditions be expected, then on closer investigation the cause will show itself to be that the elements in question are each of them members of other, fixed associative relations from which they cannot be released. It will in any case always, psychologically regarded, stand as a task to find the preceding associations which hinder later associations that are in and for themselves "natural". By isolating the two concepts element and association from one another one will thus get both a false psychology and a false physiology. For an exhaustive definition of a concept like association long investigations will still be needed, and physiology will have to wait a long time if it means to "replace" a psychological reality and not a mere abstraction.

But let us suppose that the psychological definitions are finished and complete —

[16] FLECHSIG: *Gehirn und Seele* 2. Leipzig 1896, p. 24 f.

[17] Compare the criticism of Flechsig's theory in O. VOGT (*Flechsigs Associationscentrenlehre, ihre Anhänger und Gegner.* Zeitschrift für Hypnotismus. V, 6) and ALB. ADAMKIEWICZ (*Die Grosshirnrinde als Organ der Seele.* Wiesbaden 1902, p. 75 f.).

will not the greatest problems still remain unsolved? How can physiological states have psychical symptoms? How can qualitative differences correspond to quantitative ones, and how can discontinuous phenomena be attached to continuous processes? Far from such questions falling away with the reduction of psychology to physiology, they are precisely sharpened by it and appear in a more glaring form. And in any case it stands as an unsolved riddle how the qualitative differences and the discontinuity arise. Even if one succeeds in showing that two concepts *A* and *B*, which one has hitherto regarded as different, are identical, so that one can set $A = B$, one has not thereby explained how *A* and *B* could stand as different in the first place. To call this difference merely "subjective" does not help; it is not thereby got out of existence — it is precisely the fact that existence has a subjective side as well that makes psychology possible and necessary. And if one declares (with Münsterberg) that these subjective phenomena can neither be described nor explained,[18] it is not easy to understand how one is to get anything to put into a "causal equation". One then behaves like the man who saws through the branch he is sitting on.

It is inconsistent to deny a causal relation between psychical phenomena among themselves while assuming a causal relation between the corresponding physiological states. If one acknowledges a physiological lawfulness, one must acknowledge a psychological lawfulness also. If the psychical phenomenon α corresponds to the brain-state *A*, the psychical phenomenon β to the brain-state *B*, and if there is a causal relation between *A* and *B*, then a causal relation — in the sense of unavoidable succession — must appear in psychological experience between α and β as well. And — as has been shown above — it is in reality this causal relation between α and β that leads one to look for a causal relation between *A* and *B*. In Avenarius's strict carrying through of the reduction of psychology (the doctrine of "the dependent vital series") to physiology (the doctrine of "the independent vital series") it is likewise plain at every point that it is from the "dependent vital series" that he infers the constitution of the "independent" one. For the most part what is constructed is only a physiological schematism, offered as an "intuitive" presentation of what goes on in the psychical processes. Neither physiology nor psychology is yet so complete that one can get beyond such a schematism.

The psychical phenomena do not, however, present so great a discontinuity and such purely qualitative differences as is often supposed. Exact observation often leads to the discovery that where there seemed to be a psychically empty space there was in reality a psychical content given, although it was not grasped by attention or by recognition,

[18] Münsterberg: *Psychology and Life*. p. 172: "The reality of the will and feeling and judgment do not belong to the describable world." — p. 208: "The subjective attitude is never object; it is never perceived."

or although it was forgotten immediately after being experienced. And as regards the qualitative differences, here too more exact observation will find more and finer nuances than those at which one first stopped and which one perhaps confidently declared to be wholly disparate. The continuity in conscious life can therefore be carried further than is often assumed, and there can be a scientific danger in proclaiming too strongly that difference of quality and discontinuity are the distinguishing mark of conscious life. The inner connection in the content of consciousness is a fact as well, and functions such as memory and comparison are quite as significant for the characterization of conscious life as are the phenomena that bear the stamp of discontinuity. Psychology's task therefore naturally becomes that of exhibiting, so far as possible, the connection and combination of the single elements, so that the whole is understood through the parts and the parts through their relation to the whole. That this leads to an antinomy which in the end makes the psychological problem insoluble for us, we have already seen. But there is work enough to do before we come to that limit (which moreover cannot be proved to be absolute). Leibniz has the merit of having energetically maintained the principle of continuity in the psychical as well as in the physical domain, especially by pointing to those psychical elements which — in relation to the clearly conscious states — are vanishingly small, and which only by being summed and combined bring forth a result accessible to unpractised self-observation. A comparison between modern and ancient or medieval poetry alone shows how many more mental nuances self-observation has now caught sight of; it is hardly only because differentiation in the psychical domain is now greater than in earlier times that this greater richness is found. Knowledge and understanding of what is immediate and involuntary in mental life has increased above all. Childlike and primitive psychology has a tendency to conceive everything that goes on in the soul as clearly conscious and as resting on deliberation and intention. In modern psychology the burden of proof — in practice as in theory — has been shifted: it now falls on the one who asserts that an action has been performed with deliberation and express decision. And the transition between the unconscious and the conscious, and within the conscious between the involuntary and the voluntary, itself takes place continuously; the more practised self-observation and criticism are, the more difficult it becomes to lay a finger on one definite point at which the boundary is supposed to lie.

Leibniz was inclined to reduce all psychical differences to differences of degree in respect of clearness and obscurity.[19] This attempt was characteristic of the intellectualism of the age of Enlightenment. But there are other ways than such a reduction in which

[19] *Den nyere Filosofis Historie* I, p. 334–336.

empirical psychology can conceive a continuous connection despite the qualitative differences which the psychical processes present at their different stages. Fusions and compositions take place at the different levels of conscious life, from the simplest sensations to the highest states of feeling. And displacements of motive take place, by which what at first has only mediate value later acquires immediate value, or conversely. Finally, conscious life in wholly integrated characters — psychology's highest and noblest objects — shows such a dependence of everything that stirs in the soul upon one purpose, one guiding thought, that we here find a causal connection quite as firm and as intimate as any in the physical domain. Every single thought, every single mood and every single impulse is in such a character plainly determined by the whole to which it belongs, and which it itself helps to form.

Cases enough remain in which psychical continuity cannot be exhibited. The question is whether we are therefore entitled to deny it. Apart from what continued observation may establish, the great question arises here whether psychical phenomena can come to be from material causes, when the concept of the material is conceived in the way natural science has hitherto conceived it. If the natural-scientific concept of the material includes no possibility of an arising of psychical phenomena, then it will be fully legitimate to set up the concept of potential psychical energy, in order to mark that we do not mean to let go of the principle of continuity even where we cannot apply it in detail in experience. It is indeed for a similar reason that the concept of potential energy has been introduced in physics, despite the obscurity that attaches to it. Everywhere it contains an acknowledgment of a limit at which one nevertheless does not mean to give up the connection one has hitherto found. The mere fact that psychical elements can be reproduced after an interval in which they have not been in consciousness already necessitates the use of concepts like "disposition", "trace", "possibility" and the like, which in reality express the same thing as is meant by "potential energy". That in applying this concept in the physical domain we are more happily placed than in the psychical[20] does not exclude the legitimacy and the necessity of setting it up here. One can of course, in a purely descriptive presentation, content oneself with stating the different psychical phenomena that emerge at certain moments, often without observation finding any psychical connection between them. The psychical phenomena then stand as scattered flashes, in strong contrast to the continuous connection which the corresponding physiological phenomena present. And since what appears with the highest degree of continuity easily comes to stand for us as more real than what is

[20] Compare my *Psykologi* V B, 6.

discontinuous and momentary, the physiological phenomena will easily come to stand as what is properly real, the phenomena proper, and the psychical phenomena will come to stand as something superfluous, an accidental plus, or, as it has been expressed, as epiphenomena. Such an "epiphenomenalism" has surely never been set up as a theory proper; it is merely an empirical registering of what is enigmatic in the appearance of the psychical phenomena as compared with the more closed continuity in the series of material phenomena. Nor indeed does it furnish any explanation; on the contrary it abolishes both every psychological and every physiological explanation, if it means to be more than a mere description. And even as description it easily becomes precipitate, since it easily comes to stop too early at what is discontinuous, without investigating whether there are not more transitions and nuances to be found than have hitherto shown themselves. The concept of potential psychical energy will here — precisely by being really only the expression of an unsolved task — contain a standing summons to go further in observation and analysis, in order if possible to discover the reality in which the "possibility" in the end always consists.

3. When weight is laid, from the psychological and the physiological side alike, on preserving continuity in as high a degree as possible, the identity hypothesis becomes the real working hypothesis for the problem of the psychical and the physiological. Like every significant hypothesis it is really only the expression of a method. We have in the psychical and the physiological phenomena two series of states, or forms of state, which empirically vary in a certain relation to one another, without its being possible, however, to *derive* the *existence* of the one series from the *existence* of the other. Scientifically regarded, the task must then be to conceive the two series each for itself as completely and as continuously as possible, and to show which determinate members of the one series correspond to determinate members of the other. We may rightly regard members of the one series as symptoms of members of the other. Sometimes it is the psychical, sometimes the physiological states that are the more accessible, and from which we therefore take our point of departure. The close belonging-together of the two series of states makes it impossible to refer them to two different "beings" or "things"; it will be natural to conceive them as different forms of expression of one and the same "being". From the fact that two properties cannot be derived from one another one has no right to infer that they belong to two different things, especially when they vary in a determinate mutual relation. Sulphur's form of state (solid, liquid and so on) and its colour (yellow, brown and so on) vary in a determinate relation to one another and yet cannot be derived from one another, although one can, when one knows the two series of changes, infer from the one to the other. In the case of sulphur we do know the common cause of the two series of changes, namely heat; it is heat that expresses itself in these two different ways. In the relation between the psychical and the physiological series the analogous knowledge is denied us. Our experience is here insufficient to solve the problem. This hangs together with the fact that both our knowledge of the psychical and our knowledge of the physical are empirically limited. What matters, then, is to work in both fields of knowledge in such a way that at no point does one arbitrarily cut the thread.

The physiological continuity is both a consequence of the law of the conservation of energy (really already of the principle of inertia, on the strict interpretation of it) and an expression of physiology's independence as a science. It furnishes a fruitful and indispensable principle of research by requiring constant demonstration of physiological causes of, and physiological effects of, physiological states. As we have seen above, the principle we stand at here is so essential that — if an experiment pointed in the opposite direction — one would sooner assume an error in the experiment than a breach of the principle. The principle of continuity — and with it the identity hypothesis — would be refuted if it could be shown that the energy contained in a brain-state did not stand in a relation of equivalence to preceding and succeeding states in the brain, in the organism, and in the physical world outside. It would already be a matter of concern for the identity hypothesis if it could be shown that the psychical phenomena in time go *before* or follow *after* the corresponding physiological states. But in an attempt to establish this it is not enough to show what popular observation already shows, that sensation arises later than the corresponding sense-impression, or earlier than the physiological (motor, and especially vasomotor) expressions which in one way or another hang together with it. For between the psychical states and the peripheral processes in the sense-organs and the organs of movement lie the central physiological processes, which are not so readily accessible to exact and direct experiment. And even if such a direct experiment were possible, it would not be decisive: because the phenomenon *B* follows *A* in time, *A* and *B* may very well be effects of the same cause, which first deposits the effect *A* and then the effect *B*. There might be peculiar causes which made the one vital expression alter more rapidly than the other. — Finally it would be a proof against the identity hypothesis if it could be established that different psychical phenomena could correspond to one and the same physiological state. It has been supposed, for example, that the brain's significance for conscious life might consist in counteracting organic life's tendency to habitual, automatic operation. The brain has a wonderful capacity for initiating new movements which, as soon as they have been practised, become habitual actions that are left to lower centres to carry out. Only by the brain's help has thought been able to free itself from automatism and to use the organism in its service without sinking under the power of habit. But different thoughts can at different times make use of the same "motor schema", the same suspension of an automatic tendency. This possibility has in the most recent period been advanced by BERGSON as an objection to "strict parallelism".[21] This objection touches on a question that has come up very little in the frequent discussions of the psychophysical problem. Whether or not one accepts the special theory the French philosopher has set up, it seems clear that the closer one presses the problem to life, the more extraordinarily difficult it will become to find those members of the two series of phenomena that can be said to be "corresponding". Where a qualitative difference is present, one can establish only by roundabout ways which members correspond to one another. When we know how heat acts on a substance's colour and on its form separately, we can determine which changes of colour and of form "correspond to one another". But it is precisely an analogous knowledge that is not at our disposal in the discussion of the psychophysical problem. We are then limited to purely empirical inferences. And such will here be more difficult to draw than Bergson seems to assume. For when sufficient weight is laid on continuity in the psychological and the physiological domain alike, it will prove difficult to resolve the two series into members that appear with such independence and distinctness that a real comparison can take place. But whatever hypothesis we take as our basis, we must be prepared for it that the members of the two series which may be taken to correspond to one another will present differences that could not be inferred from the one member or the other *taken by itself alone.* It might therefore very well be that "different" psychical phenomena could correspond to "the same" physiological state (or conversely), just as one language sometimes has only one word where another has two. It is then the connection that is decisive, and it will surely be decisive to such a degree — in the psychological as in the physiological domain — that it is only with a slight degree of approximation that one will be able to speak of "the same" phenomena or states. "Parallelism" is therefore not touched by that objection, provided we make sufficiently strict demands on the demonstration that the states really are the same.

It is essentially as a working hypothesis, not as a positive solution, that the identity

[21] See the interesting discussion of *Le parallélisme psychophysique et la Métaphysique positive* in Bulletin de la société française de Philosophie. Juin 1901. In the course of that discussion Bergson remarked, among other things (p. 51): *Étant donné un état psychologique, la partie jouable de cet état, celle qui se traduirait par une attitude du corps ou par des actions du corps, est représentée dans le cerveau: le reste en est indépendant et n'a pas d'équivalent cérébral. De sorte qu'à un même état cérébral donné peuvent correspondre bien des états psychologiques différents, mais non pas des états quelconques. Ce sont des états psychologiques qui ont tous en commun le même* "schéma moteur".

hypothesis (for which "parallelism" and similar expressions are unhappy and misleading designations) has its significance. I for my part have in any case always maintained it as an "empirical formula" which can guide us in our investigations in such a way that neither physiology's right nor psychology's is violated by an investigation on the one side or the other being broken off too early. The physiologist may be tempted to stop too early if he can expect at some point or other to come upon "the soul" as the cause of a change in the brain's state,[22] and the psychologist may be tempted to stop too early if he can expect at some point or other to stand before a psychical phenomenon supposed to have its cause in a "nerve process" or in "nerve energy".

4. It will nevertheless have its interest, from both sides, the psychological and the physiological, that they should approach one another as closely as possible by going back to the fundamental concepts of the two sides under which existence here appears in our experience.

From the psychological side the *concept of will*, taken in the widest sense as the concept of psychical activity, will stand as the fundamental concept. It might seem to tell against this that several psychological writers, precisely in the most recent period, seek to eliminate the concept of will from psychology — not because they deny what popular usage calls will, but because they hold that this concept does not designate so fundamental a point of view as knowledge and feeling do, the so-called phenomena of will being supposedly reducible to elements of knowledge and of feeling in peculiar combinations. In support of this one may appeal to the fact that will as such, our activity as conscious beings, cannot be an object of immediate self-observation in the way that representations and feelings can be.[23] We experience will's motives and will's results, but not will itself — just as in the domain of material nature we experience energy's conditions and expressions, but not energy itself. Hume has already shown this for all causality, and so for psychical and material causality alike. The concept of will is formed, like the concept of energy, by a construction. But it is a construction we are compelled to make. And if one determines the concept of a psychical element, not in such a way that it must necessarily be something that can be an object of immediate self-observation, but in such a way that it designates an essential and irreducible property of conscious life, then will may very well be a psychical element, and the concept of

[22] An example of this I have adduced in my *Psykologi*, 4. Udg., p. 67 Note.

[23] Compare my *Psykologi* VII B, 4; *Psykologiske Undersøgelser*, p. 67–81; *Søren Kierkegaard som Filosof*, p. 74–81. — Ebbinghaus: *Grundzüge der Psychologie* I, p. 168 (compare also his article in Zeitschrift für Psychologie IX, p. 201), and Ehrenfels: *Werttheorie* I, p. 245–249, infer from the unobservability of will that will is not a special psychical element. Compare my criticism of the last-named work in Göttinger gel. Anzeigen. 1900, p. 742 f.

will a fundamental psychological concept. The reason we cannot make will the object of self-observation, as sensations, representations and feelings can be, might lie in its extending through all the changing states and forms of conscious life as their constant presupposition. Consciousness subsists only through an unbroken labour of gathering the single elements into a whole. Such a labour of combination and concentration expresses itself in the simplest sensation as well as in every process of representation, every feeling, every impulse and every decision. At every point there expresses itself an activity which is just as original a side of conscious life as are the elements (sides or properties) that observation and analysis immediately find before them. It is not the case that we first have sensations, representations and feeling, and that then, by their combination, something arises which we can call will. Without an original combination, a primary synthetic process, the very elements which determine will in the narrower sense do not arise at all. In the special expressions of will (reflex action, impulse, wish, intention, decision) the primary capacity of concentration expresses itself in a peculiar way under the influence of certain determinate elements of knowledge and of feeling.[24] — There thus takes place a constant interaction between the element of activity, the will-element proper, and the intellectual and emotional elements. We meet here again the antinomy mentioned above (1), which asserts itself in all conscious life, indeed in all life. But the main thing here is that we can form a concept of energy already on purely psychological ground, since psychical work is performed wherever a psychical phenomenon appears, this always — so far as we can trace it — presupposing a synthesis. The psychical work in which the synthesis consists is the greater, the more the single elements are qualitatively different, and the further apart they lie in time.

It might now be tempting to identify this psychical energy straightway with the energy that operates in the nerve tissue. But since not all nerve processes are connected with phenomena of consciousness, a distinction must be drawn between conscious and unconscious nerve energy, and the problem poses itself anew. Besides, we know nothing more precise about this so-called nerve energy, or about its relation to other forms of energy, organic and inorganic. In taking the riddle to be solved — as Ostwald[25] and others do — by the very setting up of the concept "nerve energy", one merely introduces a

[24]See more on this in my *Psykologi* II, 5; IV, 7 e; V B, 5; VII A, 1; B, 4–5.

[25]Ostwald seeks in his *Naturphilosophie* (Leipzig 1902) to show that the concept of energy is the fundamental natural-scientific concept, and since the phenomena of consciousness too must (with Kant) be conceived as expressions of activity (*Bethätigungen*), it becomes the fundamental psychological concept as well. But about the relation between the two kinds of energy he does not express himself clearly. Now the processes of consciousness are themselves called "energetic" (p. 394), now consciousness is said to be conditioned by the nervous system's energy (p. 396), now spiritual energy is defined as "conscious *or* unconscious nerve energy" (p. 398).

qualitas occulta and acquiesces in it. The natural-scientific concept of energy is, wherever one meets it, always abstracted from phenomena with geometrical properties. Natural science knows energy only as an expression of the relation between phenomena in space. The riddle would therefore be solved only if we could form a concept of energy from which the psychological and the natural-scientific concepts of energy could be derived as special forms; but we lack the means to construct such a concept. In any case every attempt in that direction would carry us beyond the domain of the psychological problem.

II. The Problem of Knowledge

1. Not only psychological understanding, but all understanding, is conditioned by the relation between continuity and discontinuity. The latter conditions the fullness and the manifoldness of understanding's content; the former conditions its gathering-together and its ordering. Understanding appears under different principal forms, to which different kinds of science correspond.

I understand what something is when I recognize it again. Thus I understand, or know, who is coming when I recognize the one who is coming. Recognition rests on there being a connection between new and earlier experiences. In the act of recognition all intervening experiences are pushed aside, and the new phenomenon is identified — immediately or mediately, involuntarily or after some deliberation[26] — with a phenomenon that has occurred earlier. The contrast with what is new co-operates here as well; the quality of familiarity becomes the more strongly marked, the more unfamiliar the surroundings are. — It is recognition that makes *description* and *classification* possible; though the apprehension of difference co-operates essentially in both operations. In all describing and classifying certain types stand fast, at least provisionally fast, and the new phenomena are referred to them, either as entirely the same (coinciding with one another), or as similar (qualitatively alike), or as presenting analogy (likeness of relation). The positive concepts we form express such types. Where we can have recognition of no degree at all, and so can set up no type, we provisionally collect the phenomena under a negative concept — that is, we give them the common mark of being different from the types hitherto determined. The negative concepts have this significance in the history of classification, that by their help one gathers what has hitherto been undetermined into a group distinct from what is determined, so as then to leave it to later research to find positive determinations within it.[27]

In the history of philosophy Plato's doctrine of Ideas stands as the characteristic expression of the significance of describing and concept-forming inquiry. The very

[26] On the different kinds of recognition see my *Psykologiske Undersøgelser*, p. 35 f.

[27] See *Det psykologiske Grundlag for logiske Domme*, p. 32 f.

possibility of recognition amid the motley manifoldness of the phenomena fills Plato with enthusiasm, and the highest spiritual capacity was for him that by which the different types underlying recognition are ordered according to their mutual likeness and difference, so that a world of thought is formed within which one can ascend and descend by steps that form a continuous series.

When concepts have been formed in different connections, they can be combined in judgments, and when different judgments have concepts in common that can be substituted for one another, inferences can be drawn. By the road of *inference* concepts and judgments can be formed without one's needing to go back to experience, as the describing and classifying sciences are obliged to do at every point. A continuity of a higher kind here asserts itself, in that thoughts and series of thoughts of the most various origin can be brought into connection with one another. While recognition rests merely on identity, inference rests on rationality: it is the relation between ground and consequent that governs, and that makes a new kind of understanding possible. I understand $A = C$ when I know $A = B$ and $B = C$. This kind of understanding is won through the formal sciences. It is peculiar to them that in the end it is a matter of indifference whence the first concepts and judgments derive, provided only they are of such a character that chains of inference can be built upon them. Just so, the describing sciences do not trouble themselves about where the phenomena come from, provided only they can be ordered according to relations of likeness and difference.

A third kind of understanding is characterized by this, that one neither merely ascends from phenomena to concepts, nor merely forms by inference new combinations of concepts out of given combinations of concepts, but derives new phenomena from phenomena given earlier. The new phenomena are understood by being set in a relation to earlier phenomena analogous to that which subsists between ground and consequent in an inference. This kind of understanding is characteristic of the exact real science (natural science), as it has developed especially since the age of the Renaissance. It is here the *concept of cause* that governs. We here connect experiences according to their unavoidable and law-determined succession in time.

This third kind of understanding has especially occupied the epistemology of more recent times, after Hume and Kant had impressed the difference between logical inference and real causal explanation. The possibility of forming and applying the concept of cause has in more recent times awakened a wonder similar to that which the possibility of forming general concepts awakened in Plato. It is, however, not only the intellectual need to find connection among experiences that has led to placing the concept of cause first, but also the need to distinguish sharply between subjective representations and objective reality, since in doubtful cases the criterion of reality is always in the end the firm, unbreakable connection.[28] The world of dreams and of thoughts is larger than the world of reality (Eng ist die Welt, und das Gehirn ist weit), and men have felt ever more the need to get their thoughts so determined and so connected that they can stand as the expression of a reality — especially since only thereby can they hope to find the means of attaining their ends.

The concept of cause appears in two forms: a provisional, elementary form at which we are often compelled to stop, and an ideal form towards which all inquiry and all theories strive. *The elementary concept of cause* indicates merely an unavoidable relation of succession: when the phenomenon *A* has occurred, the phenomenon *B* thereupon unavoidably occurs, and *B* occurs only when *A* goes before. It is not thereby said that the causal relation obtains between *A* and *B* themselves; it is possible that both are successively appearing effects of an antecedent cause. *The ideal concept of cause* goes a step further and sees in the phenomenon called the effect a continuation of, or a conversion into new form of, the phenomenon called the cause. The ideal concept of cause thus passes over into the concept of development, wherefore it is no wonder that this concept, alongside the concept of cause, plays so great a part in the science of more recent times.

2. All three kinds of understanding rest on certain fundamental propositions or principles which involuntary thought-work — as it is practised by practical common sense, in the special sciences, in philosophical speculation and in religious thinking — feels no need to seek out, but which epistemology brings forward and tests. The problem of knowledge arises when it is asked what *the validity of understanding* rests on, and how far it extends. It is the relation between continuity and discontinuity that here again poses the problem. For Plato the problem arose from the irrational relation between concept and phenomenon; in more recent times, from the irrational relation between formal and real science. Can the concept ever be an adequate expression for the manifoldness of the phenomena, and the law for their changing combinations?

For Plato the consequence was that the concepts ("the Ideas") must have been known to man from a higher existence, and now emerged for thought in their whole perfection under the influence of the imperfect imitations of them which the world of sense presents. In more recent times the thought of nature's lawfulness has often been taken for an a priori truth, an original intuition which at most owed to experience the

[28] See *Det psykol. Grundlag*, Kap. 6.

occasion of its being called forth. It is the incommensurability between the principles (the logical principles and the principle of causality) and experience that has given birth to this speculative epistemology. This incommensurability has also given birth to a theory one might call the arbitrary theory, because it strongly stresses what is arbitrary in the first setting up of the principles — but which stresses no less strongly that the principles in their pure form can win no confirmation from experience, can never become results, but always only postulates. There can — says HOBBES, the most pronounced representative of this standpoint — be no science of the very principles of all science; they are set up by constructive art (*principia sunt artis sive constructionis, non autem scientiæ et demonstrationis*), and it is therefore we ourselves who bring forth their truth (*rationis prima principia vera esse facimus nosmet ipsi*). In FICHTE the first and unconditioned fundamental proposition of all human knowledge is set up by a free act; our science rests not on a "Thatsache" but on a "Thathandlung", namely the thinking consciousness's resolve always to remain in agreement with itself: the setting up of the principle of identity. Similarly S. KIERKEGAARD assumes a leap in the setting up of the principles, and KROMAN derives the principle of identity from the instinct of self-preservation, which seeks to maintain consciousness's unity; "from experience it has in no way arisen".[29]

Both the speculative and the arbitrary theory must, however, concede that there must be determinate empirical occasions which call forth the intuitions or the postulates. Empiricism (which — when it takes account not merely of the single individual's experiences but of the whole race's — passes over into evolutionism) now lays, where it unfolds in its absolute form, the chief weight on these "occasions" and regards them as the complete causes. It is most clearly presented by STUART MILL and HERBERT SPENCER. It seeks to show how general principles may have formed themselves successively in consciousness under the constant influence of the surroundings, and it energetically maintains that no principle can have any validity beyond the confirmation by experience (verification) that it can win. Only as results, then, have the principles value.

What empiricism cannot explain is the fact that the principles themselves call forth experiences, through the questions they lead us to ask. More happily placed here is a fourth theory, developed in the most recent period by ERNST MACH and RICHARD AVENARIUS, which one might call the economic. It lets both passive experience and

[29]HOBBES: *Logica.* Cap. 3 (§§ 8–9); *Physica.* Cap. 25 (§ 1). (Elsewhere, however, Hobbes teaches that it is analysis proceeding from what is given that leads us to the principles: *Analytica est ars ratiocinandi a supposito ad principia, id est ad propositiones primas vel ex primis demonstratas, qvot sufficiunt ad suppositi veritatem vel falsitatem demonstrandam. De rationibus motuum et magnitudinum.* Cap. 20 (§ 6).) — FICHTE: *Grundlage der gesammten Wissenschaftslehre.* § 1. — S. KIERKEGAARD: *Uvidenskabelig Efterskrift*, p. 83. — KROMAN: *Vor Naturerkendelse*, p. 270–274.

active development of thought come into their right, in that it regards the principles as those "conceptual reactions" which by the shortest road make intuitive apprehension and survey possible. When the principle of continuity plays so great a part, it is because it is a decidedly economic principle. There is here, then, only a question of practical applicability. One forms the concepts and principles one needs in order to make oneself master of the phenomena. Every practical and intellectual need is satisfied when our thoughts are fully able to reproduce the sensible facts. This reproduction is physics's end and aim, and atoms, forces and laws are only means of easing the reproduction. Their value reaches only so far as they help us.[30] Maxwell and Hertz have expressed themselves in a similar direction.[31]

The four theories, which respectively conceive the principles of knowledge as intuitions, postulates, generalizations and economic instruments of thought, all presuppose *the analytic or regressive epistemology* which Kant in particular has developed. For *what* is to be the object of intuition, or of postulate, or of generalization, and *what* answers to the economic needs of research, can be discovered only by *inferring back from the data which experience presents, and finding the presuppositions on which the understanding of them rests.* Such an analysis must lie at the ground of every epistemology. Whether this analysis has been completed, one will not be able to convince oneself with full certainty. The history of science shows that here is a labour that must be taken up again and again. Now one supposes one needs more, now fewer principles than hitherto. A guarantee that one has reached the absolutely last presuppositions one will never be able to obtain. What the economic theory — to which I shall chiefly hold — particularly stresses is, partly that no more principles are to be set up than what is given demands with strict necessity (this is what is meant by "critique of pure experience"), partly that at different times, in different scientific situations, different principles may be needed, so that even a principle which has for a long time carried research forward may later be recognized as invalid, without its historical significance being thereby misjudged. The economic theory thus stresses partly the law of parsimony, partly the purposive, utility-determined character of the principles. It is due especially to two classes of motive: first, the wish to reduce to the least possible those presuppositions that cannot themselves be proved — an antidogmatic or antimetaphysical motive; and secondly, experiences from the history of the sciences, which show how principles and hypotheses may be legitimate and fruitful

[30] Ernst Mach: *Beiträge zur Analyse der Empfindungen.* p. 144.

[31] Maxwell: Scientific Papers. II, p. 360. — H. Hertz: *Einleitung.* (Werke III, p. 1 f.). — Compare an interesting discussion *De la valeur des lois physiques* in Bulletin de la société française de philosophie. Mai 1901.

in a certain period but must later be replaced by others — something to which the most recent period's discussion of the grounding and the validity of the mechanical view of nature has particularly drawn attention.

There are, however, two sides of the matter which the special form the analytic epistemology has assumed in appearing as economic theory has not sufficiently stressed, and which become of particular importance when one does not regard the problem of knowledge in isolation but sees it in its connection with the other philosophical problems.

In the first place, the forms in which the intellectual need is satisfied must hang together with the general nature of consciousness. *What* we understand, and *that* we understand anything at all, depends not merely on the constitution of the phenomena but on our intellectual organization, just as which colours we see depends on the constitution of our organ of sight and not merely on the outward objects. There will be a certain type for all principles and hypotheses, one which points back in the end to the innermost nature of conscious life. And it will here always be the need for unity and continuity that one comes back to. Which and how many fundamental concepts (categories) and fundamental presuppositions one must set up is a problem that must always be taken up anew in knowledge's struggle to press forward. It was an illusion when Kant supposed that one could give, once for all, a catalogue of what is needed in this respect; but the old master was not mistaken in this, that it is *the need for unity and continuity* that lies at the ground of all the forms in which understanding is won, or is supposed to have been won. He has himself shown that all his categories can be traced back to the concept of continuity, and through all vicissitudes in the domain of principles it will undoubtedly be this concept that constantly recurs. The logical principles, the principle of causality and the fundamental propositions of natural science all lead back to this concept, which hangs so closely together with the nature of conscious life. A purely psychological epistemology will, however, never be able to suffice; for from the fact that the need for unity and continuity — be it ever so essential to conscious life — is satisfied by certain principles, it by no means follows that these are objectively valid. That need can find vent and satisfaction in many ways and in many forms, as the history of mythologies and of speculations sufficiently establishes — ways and forms which as a rule by no means satisfy the demands of economy, whether in respect of parsimony or in respect of applicability. Here Schiller's words hold: Eng ist die Welt, und das Gehirn ist weit. Of all the wealth of thought at the human spirit's disposal, only a narrow and firmly closed series of thoughts can be used when it is a question of valid understanding. What is a necessary presupposition is only this, that the presuppositions which the understanding of what is given requires must be psychologically possible, must accord with the general laws of conscious life, and be only more special and more exact developments of what lies in those laws. A selection must take place among the thoughts that involuntarily press forward; but it brings with it no emancipation from the general psychological laws. What matters is that new dispositions of a special kind should form themselves — that there should arise an intellectual habitus which poses the questions and criticizes the answers in a stricter and more determinate way than the involuntary course of thought feels any need to do. Every comprehensive principle is really — psychologically regarded — the expression of such a habitus, which may be more or less deeply grounded in the nature of consciousness, that is, may now have the character of an instinct, now be due to the influence of practice. The purely logical principles come closest to the instinctive. The need for agreement with oneself, for consistency in the course of representations, cannot be explained from parsimony and expediency alone. Where strict inference from the experiences so far available leads us to self-contradictory results, we far sooner assume that the experiences are incomplete than that existence should contradict itself. What holds in the highest degree of the purely logical principles holds of more special principles as well. Thus the principle of causality expresses a tendency, when an event has occurred, to look about for other events in which the conditions of its occurrence might be found. Here too a need asserts itself to bring unity and connection into the content of consciousness. The more special and determinate the satisfaction of this need is to be, the more the economic principle operates in its two forms: as parsimony, which takes the straight road to the goal, and as expediency, which follows roads that really lead to understanding. Examples of this are to be had in Kepler's and Newton's demand for *veræ causæ*, in the principle of inertia, in the principle of energy, and so on. What purely empirically stands as a hypothesis becomes, epistemologically regarded, a principle, a guiding thought under whose leadership consciousness can, in the world of experience, satisfy its need for unity and continuity.

In the second place, principles and fundamental hypotheses — even if in the end they all stand as *working* hypotheses in the service of intellectual economy — must not be conceived as wholly accidental or arbitrary in relation to the existence which we mean to understand by their help. The very concept of a working hypothesis points in two directions: partly — as already shown — back to the nature of the thinking consciousness, since our consciousness cannot carry out a function, be it ever so economical, that is wholly foreign to its own essence; and partly outward, towards the existence to which the phenomena that are to be understood belong. A working tool must fit both the hand that is to use it and the object that is to be worked upon. Existence must then in itself present sides that answer to the unavoidable presuppositions of our knowledge, however much these are conditioned by the inner and outer circumstances under which knowledge works — or perhaps precisely for that reason. This holds for all valid knowledge, from the most elementary forms to the highest. It holds already of the sense-qualities, despite their "subjectivity", and it holds of the highest principles of unity in our thinking. KANT says of the tendency to presuppose a unity behind the differences of natural phenomena:[32] "One might perhaps believe that we had here a merely economical contrivance of reason, serving to spare effort .... But such an egoistic purpose is very easily distinguished from the Idea according to which everyone presupposes that this unity of reason accords with nature itself, and that reason here does not beg but commands, although it cannot determine the limits of this unity." I add, however: even where reason commands (or asks, expects, anticipates, postulates), it is obliged — as all commanders are — to adjust its commands to the capacity for obedience that may be presumed to be present. —

This last consideration leads in particular to making possible the connection between the problem of knowledge on the one side and the problem of existence on the other. It is all the more important to bring this out, since the analytic epistemology — like the economic, and in its way the arbitrary as well — leads to putting a new concept of truth in place of the current, naive one. The significance of the principles is to guide our work of winning understanding. Their *truth* consists in their *validity*, and their validity in their *working value*. That a principle is true means that one can work with it; and that means, when the principles of knowledge are in question, that by its help one can get forward in understanding — in firm ordering and connection of the phenomena. The concept of truth is a *dynamic* concept, in that it expresses a determinate way in which the energy of thought is employed, and a *symbolic* concept, in that it does not mean coincidence with, or qualitative likeness to, an absolute object, but likeness of relation (analogy) between the events in existence and human thoughts. The old naive concept of truth, according to which a piece of knowledge was true when it directly pictured or mirrored "reality", is untenable, and has been so ever since the moment when men began to assert *the subjectivity of the sense-qualities.* That the sense-qualities were subjective was not, after all, to mean that they were invalid, unusable for orienting us in the world. They still stand as signs, signals, symbols, whose mutual succession we can interpret as the expression of an objective series of events, although they cannot be proved to be pictures of it. It stands similarly with the logical principles and the other

[32] *Kritik der reinen Vernunft.* p. 653.

fundamental presuppositions of our knowledge. *Critical philosophy* led to the result — a simple consequence of the very analytic method it employed — that the truth of the fundamental propositions can mean only that they make scientific experience possible; they were themselves, after all, found by analysis as the necessary presuppositions of such experience. A comparison between our thoughts and an absolute world of things is not possible; we can compare only thoughts and experiences with one another. This consequence Kant himself did not see so clearly as some of his disciples did (Maimon and Fries);[33] the master himself was still entangled in dogmatism, as is seen from his doctrine of the "Ding an sich" as absolutely subsisting outside every subject. But when he designates the law of causality an "analogy of experience", meaning thereby that events in time stand to one another as ground and consequent do in our thinking, he comes close to making it a working hypothesis. In the most recent period one is — as we shall soon see — *on the side of natural science* more inclined to acknowledge the dynamic and symbolic concept of truth than one was so long as the mechanical view of nature stood with a certain dogmatic character. — This new concept of truth, which is pressing forward in the domain of science, presents a kinship with the way in which — as we shall see under the fourth principal problem — the religious consciousness more and more places itself over against the dogmas; in the religious domain too it is more and more applicability and working value that are asked after. The static concept of truth[34] is everywhere being pushed aside by the dynamic.

Even supposing that fruitful principles or working hypotheses have been won, will existence then be exhaustively reproducible by their help, or will there always continue to subsist *an irrational relation between the principles our consciousness is able to form and existence itself, from which our experiences derive*? — It will appear that under three forms especially something irrational will always remain: in the relation between quality and quantity, in the significance the time-relation has for the concept of cause, and in the relation between subject and object. We shall consider these three points each by itself.

3. In the attempts to reduce all given differences to identity and continuity, the endeavour to trace differences of kind back to differences of degree has been of particular importance. In the science of material nature it expresses itself as an endeavour to explain all change as motion in space. Motion from one place to another is the simplest change; it would therefore be a great advance in clarity if it could be shown to be the change which, under different forms, takes place everywhere in nature where immediate experience

[33] *Den nyere Filosofis Historie* II, p. 114 f.; 222.

[34] The expression "static concept of truth" was used — with a somewhat different motivation — by Louis Weber at the philosophical congress in Paris in 1900.

shows us qualitative changes. Aristotle already taught that spatial motion lay at the ground of all other change, and of all coming-to-be and passing-away as well; he did not, however, hold it possible — as the Atomists had wished — to trace everything in material nature back to it. Only with the rise of modern natural science does this thought come forward in earnest. Galilei stands as the great founder of what one is accustomed (in the narrowest sense) to call *the mechanical view of nature.* Besides the simplicity thereby attained — and Galilei was certain that nature follows the simplest ways — there is gained the further advantage that one can operate with determinate quantities. Galilei therefore said: measure all that is measurable, and make measurable what is not! By this reduction agreement becomes possible, whereas qualitative differences can only be estimated and not measured; and exact verification becomes possible.[35]

The principles on which the mechanical view of nature rests were regarded by its most important representatives as absolute truths, fundamental laws of existence; in so far as it was still thought that they needed a grounding, this was taken directly from God's essence and will. In this Cartesians and Newtonians were agreed. And the Materialists differed from them only in holding the theological grounding superfluous and impossible. Through a series of magnificent discoveries and explanations this view of nature has shown its fruitfulness. The question for us is whether, from an epistemological point of view, it can be said to mark an absolute conclusion.

Even supposing that everything in material nature could be explained by the principles of the mechanical view of nature — and there is, as we shall soon see, doubt of that in the most recent period within the circle of natural scientists — it is in the first place clear that the qualities are not got out of the world by being "reduced" to quantities or charged to the account of the sensing subject. They remain standing as immediate facts which must be established empirically. The chemical product's properties cannot be derived from the properties of the elements; and when one kind of physical energy receives its equivalent in another kind of physical energy, the equivalent nevertheless has new properties which cannot be derived from the properties of the first form of energy. One cannot, however, let the sensing subject create these qualities out of nothing; in any case we then get insoluble psychological difficulties in place of the physical and chemical difficulties we have pushed aside.

In the second place, extension and motion are in the end themselves qualities in the widest sense — factual properties which in and for themselves might just as well need an explanation as the specially so-called sense-qualities. Since Berkeley's and

[35] *Den nyere Filosofis Historie* I, p. 160–167.

Leibniz's time this has often been urged from the philosophical side. There is no reason to assume that the quantitative properties express the innermost essence of things. The reason science seeks them out by preference and holds to them is really only that by their help one can give exact descriptions of the material phenomena and draw determinate inferences from them. This has in the most recent period been strongly emphasized by investigators such as Maxwell and Hertz — in the latter's case with express adherence to the economic epistemology discussed above. "The progress of the exact sciences", says Maxwell, "depends on the discovery and development of appropriate and exact ideas, by whose help we can form a mental representation of the facts which is at once sufficiently comprehensive to stand for every particular case, and sufficiently exact to afford security for the inferences we draw from them by way of mathematical calculation." And Hertz says: "We form for ourselves such inner images (*Scheinbilder oder Symbole*) of outward objects that the logically necessary consequences of the images are always in turn images of the naturally necessary consequences of the objects imaged .... Any other agreement with the things, the purpose of our representations does not require."[36] — The dynamic and symbolic concept of truth is here expressly put in the place of the naively dogmatic one which the mechanical view of nature had formerly espoused. What matters is only to find a circle of symbols that can be applied with full consistency, and from which inferences can be drawn that are confirmed by new experiences, which in their turn can themselves be expressed by the same circle of symbols. But the possibility can here never be ruled out that another circle of symbols would be able to express the positive experiences equally well or better, and to serve as the ground of exact and verifiable deductions. It will never be possible to prove of any circle of symbols that it is the only right one, and the only necessary or possible one.

The epistemological reflections to which recent natural scientists have thus been led were called forth by the difficulty that presented itself in bringing the electrical phenomena under the mechanical view of nature, or else in deriving that view's principles from the laws holding for those phenomena. Hertz inclines rather to the latter possibility, Boltzmann rather to the former.[37] But of greatest interest for epistemology is Maxwell's criticism of the current concept of matter, according to which by matter one imagines an extended mass. The fault in this concept is, according to Maxwell, that it leads one

[36] See Note 30.

[37] H. Hertz: *Ueber die Beziehungen zwischen Licht und Elektrizität.* 1889, p. 29. — Boltzmann's lecture at the German congress of natural scientists 1900 (translated in The Monist. January 1901: *The recent development of method in theoretical Physics*) and his paper in the Vienna Academy's Oversigter CV, 8 (translated in The Monist. Octbr. 1901: *On the necessity of atomic theories in Physics*). — On the whole question compare also C. Christiansen: *Den elektromagnetiske Lysteori.* (Det danske Vid. Selsk.s Oversigter. 1889), and the same author in *Fysisk Tidsskrift* I, p. 4–5.

to think of matter as something at rest. But it is motion that makes rest intelligible to us, not the reverse. The doctrine of motion must therefore precede the doctrine of equilibrium, dynamics precede statics. It is from motion that we win the concept of force or energy, by whose help equilibrium becomes intelligible to us. If now by matter we understand what is constant, what subsists through all changes, then it can only be the law of motion itself, and the essence of "matter" consists accordingly in motion. It is, according to Maxwell, a prejudice to regard matter as extended and the molecules as hard and solid: one would then have to ask what holds the molecule's parts together, and one would arrive at molecules of a second order. Yet we must constantly operate both with geometrical and with dynamic concepts, whether we regard the last elements as extended or not. It is only extension as a resting property that Maxwell declares against; even with a line we draw on a board, the motion by which it comes to be is the essential thing.[38] — It is an extraordinarily important epistemological point of view that is here brought forward: that resting states always contain problems which can be solved only by rest's being replaced by motion. Potential energy we understand only through actual energy, capacities and dispositions only through what comes of them. This law holds everywhere, in the psychical domain as in the physical. There is a half mystical, half materialistic inclination to find completion in contemplation of — or staring at — a resting "something".[39] Maxwell has made an important contribution towards getting this inclination rooted out. But the great question is whether the thought of the continuity of motion, or of activity, can be carried through in all domains. Even if the dynamic asserts, epistemologically regarded, its priority over the static, the static cannot for all that be got rid of; it presents itself ever anew with its tasks for thought, and Maxwell himself acknowledges it when he declares both geometrical and dynamic concepts necessary in the explanation of nature. In contrast to the dynamic, the geometrical designates simultaneity. In the domain of material nature the simultaneous appears in the form of space;[40] in the psychical domain the relation of simultaneity does not assume this

[38]Maxwell: *Scientific Papers.* II, p. 26–33; 302–305 (where M. shows what he owes to Faraday and Lord Kelvin); 326; 777 f.

[39]Compare my *Psykologiske Undersøgelser*, p. 77–81.

[40]This point Ostwald does not let come into its own when he wants to trace everything in nature back to "energy". Mass, the filling of space, weight and chemical properties are all supposed to be traceable back to the concept of energy. It is correct that all those concepts presuppose the concept of energy; but is that the only fundamental concept in natural philosophy? In that case one would have to be able to derive the "geometrical" properties from the dynamic ones, and at that Ostwald makes no attempt. In his work *Die Ueberwindung des wissenschaftlichen Materialismus* (1895) he even defines matter as "a spatially [sic] ordered group of different energies" (p. 28). This spatial ordering, here acknowledged as a special moment, is — it seems to me — somewhat pushed aside in his *Naturphilosophie* (1902), where "the energetic world-picture" is developed. Yet his main proposition, on which he builds his system, is that "every process without exception can be exactly and exhaustively described or presented by indicating

geometrical form, but appears nonetheless as something "static" which sets research new tasks when it has happily and successfully found the laws of the psychical processes. After the continuity in the processes of development has been exhibited, it remains to exhibit that there is continuity between the processes and the resting states. — We meet here again, then, the standing problem; it would present itself even if the qualities were fully explained by being represented by quantities, and even if the fundamental physical concepts with which science operates were more than the most perfect and most expedient circle of symbols that it has so far succeeded in forming and applying. The possibility of an irrational relation between existence and our knowledge cannot therefore be ruled out.

4. The investigation of the relation between *the elementary and the ideal concept of cause* will lead us to a similar result. In its elementary form the concept of cause indicates, as we have seen, only this relation between two events: that when the one has occurred, the other unavoidably occurs. But from this unavoidable succession research advances, through observations, experiments and hypotheses, so far as possible to the exhibition of a continuity in the series of events, by which the differences between the single members of that series become vanishing, so that at last even the temporal difference between the events is reduced to the least possible. From outer, though unavoidable, succession thought works its way through continuity towards a complete identity. While the time-relation plays an essential role in the elementary concept of cause, it is as good as eliminated in the ideal concept of cause. If this process could be carried through completely, we should arrive at the paradoxical result that the perfect causal explanation would be an abolition of the concept of cause itself: for the causal relation differs from the purely logical relation between ground and consequent only by the temporal difference between the members of the relation.

The philosophy of the seventeenth century (Spinoza and Leibniz) did not distinguish between cause (causa) and ground (ratio); the causal relation between two phenomena was for them the same as the relation between premisses and conclusion in an inference, a mere relation of identity. It stood for them as self-evident that there could not be more in the effect than there was in the cause; the time-relation between the members was

which energies undergo temporal or spatial [sic] changes" (p. 152). That the concepts energy and time cannot be separated is a matter of course, since energy means the capacity to overcome a resistance, and every overcoming takes time. Not so much a matter of course is the connection between the concept of energy and spatial changes. This connection derives, in Ostwald, simply from his having abstracted the concept of energy from spatial phenomena, from investigations of changes at determinate places in space. Only by this geometrical moment's being pushed aside in his concept of energy does it become possible for him to believe that the concept is without further ado the same for psychical and for physical phenomena, so that the problem of "soul and body" would be solved by setting it up. See above, Note 24.

indifferent, and time on the whole belonged only to the obscure sphere of sensuous representation (imagination). When Hume raised the problem of causality, he did so precisely by emphasizing that no proof can be given for the justification of inferring from past to future. When appeal is made to the experiences of the past, Hume says, these can only show that a thing has at some time, in a single moment, been in possession of a certain power, not that it constantly possesses it and will possess it.[41] Kant believed indeed that he could give a rational proof of the validity of the causal principle, but he distinguished between the causal relation and the relation between ground and consequent, in such a way that there was only an analogy, not an identity, between them: the law of causality meant for him that events follow one another in time in a manner analogous to the manner in which the conclusion springs from the premisses. The causal principle therefore gives us a rule by whose help we can attain unity in our experiences.[42]

From two different standpoints attempts have been made in the most recent time to eliminate the time-relation and thereby to abolish not only the elementary concept of cause, but in the end the concept of cause altogether.

From a speculative standpoint it has been maintained that the time-relation always points to an imperfection, to an incompleteness in development. So long as the time-relation is determinative for our conception of existence, our conception is therefore always imperfect, and the reality of the time-relation is irreconcilable with the idea of a perfect knowledge, of an absolute truth. Our knowledge does in fact work itself more and more away from the time-relation; the more clearly we understand something, the smaller the role the temporal difference plays — the more does real knowledge pass over into formal knowledge, cause into ground. What we begin by calling cause and effect comes, as knowledge advances, to stand as members in a whole, members which stand in a lawlike relation to one another, a rational relation for which the temporal sequence is unessential. That we first observe the one member and thereafter the other has no significance: in our knowledge of the law the whole process stands before us as a unity. It is not the time-relation, but the unity behind the time-relation, that binds what

[41] Hume: *Treatise on human nature.* I, 3, 6: Your appeal to past experience decides nothing in the present case, and at the utmost can only prove, that that very object, which produc'd any other, was *at that very instant* endow'd with such a power.

[42] *Kritik der reinen Vernunft*: p. 181: Wir werden also durch diese Grundsätze [the causal principle and the kindred principles concerning the persistence of "substance" and the reciprocal action between all that exists] die Erscheinungen nur *nach einer Analogie mit der logischen und allgemeinen Einheit der Begriffe* zusammen zu setzen berechtigt werden. — p. 180: Eine Analogie der Erfahrung wird also nur eine Regel sein, nach welcher aus Wahrnehmungen Einheit der Erfahrung entspringen soll, und als Grundsatz von den Gegenständen (der Erscheinungen) nicht constitutiv, sondern nur regulativ gelten.

we call cause to what we call effect. Besides, neither is there any interval between the end of the event we call cause and the beginning of the event we call effect. It is the English philosophers Francis Bradley and Bernard Bosanquet[43] who have urged this consideration, at the base of which lies a speculative interest in a total conception and in concluding concepts.

From a wholly different standpoint, which may be designated the standpoint of "pure experience", and whose most important representatives are Richard Avenarius and Ernst Mach, work is done in a similar direction, but from other motives. The interest here is directed to separating out all the accessory representations and auxiliary concepts with which we, involuntarily or on methodological grounds, supplement experience itself, the immediately given. The advance of knowledge consists in a reduction of the differences (to a "heterotic minimum") and in an approximation to a pure description of a continuous process. In the course of this advance the content of knowledge is more and more restricted to descriptive statements, so far as possible with analytic transitions, and the differences are reduced from qualitative to quantitative, so far as possible with a constant exhibition of relations of equivalence. It is no longer the task of strict science to "explain"; whoever wants "explanations" is referred to mythology and metaphysics; the aim of science is to give an exact, methodical description of all relations and transitions. As regards the causal relation in particular, weight is laid partly on this, that every application of the concepts cause and effect arbitrarily draws two elements out of the whole connection in which they occur and sets them in opposition to one another; partly on this, that the time-relation may very well be reversed as soon as one has exhibited a relation of equivalence, and as soon as one abstracts from the direction of the changes.[44]

The epistemological views which thus, with different motivations, would eliminate the elementary concept of cause and deny the fundamental significance of the time-relation may be characterized by this, that they put an ideal of knowledge in the place of the actual knowledge we can attain at any given time. Every determinate investigation must begin at a determinate point, which lies where the problem presents itself. And the problem is set when two members in the series of events draw attention to themselves and awaken surmises of an inner connection between them, or when a single member

[43] Francis Bradley: *Appearance and Reality.* London 1893. Chap. 4–7; 18. — Bernard Bosanquet: *Logic or the morphology of knowledge.* Oxford 1888. Book I. Chap. 6.

[44] Rich. Avenarius: *Kritik der reinen Erfahrung.* II. p. 332–339. — Ernst Mach: *Zur Analyse der Empfindungen.* p. 168: "Die Thatsache der Nichtumkehrbarkeit der Zeit reducirt sich darauf, dass die Werthänderungen der physikalischen Grössen in einem *bestimmten Sinne* stattfinden. Von den beiden analytischen Möglichkeiten ist nur die eine wirklich. Ein metaphysisches Problem brauchen wir hierin nicht zu sehen." — Compare on these theories Heinrich Grünbaum: *Zur Kritik der modernen Causalanschauungen* (Archiv für systematische Philosophie. 1899. p. 392–409).

steps forth in its peculiarity or its suddenness and thereby awakens the need to seek intermediate members through which it can be brought into connection with the whole remaining series. It is to that extent an accidental, arbitrary starting-point that is seized upon; but from the nature of the case it cannot be otherwise. Our knowledge develops historically, because human attention is awakened only under certain determinate conditions. If it were directed without difference and at all times equally upon all members in the series of events, no knowledge would arise. It is of course of importance that one be conscious of what is accidental or subjective in the starting-point and in the section taken; and if the work of knowledge has success, one does get beyond it, since the section is precisely through understanding worked into a great continuous connection.

The same historical character of our knowledge becomes clear when we bethink ourselves that we always stand between the experiences of the past and the possibilities of the future. *In each single moment* we must distinguish between what lies given and what we expect; the former is the cause, the latter the effect; and the expectation can never become more than a hypothesis. The single moment, in which on the one side a "no longer" holds and on the other side a "not yet", sets the problem before us in its whole tension, a tension which only the power of habit diminishes. With new problems the tension will not fail to appear — and so long as knowledge goes forward, it will seek and find new problems.[45] The full connection is always present only afterwards, and until it is present the assumption of it stands as a hypothesis. We shall therefore never be able to dispense with concepts such as force, energy, cause or possibility, which with different nuances express one and the same relation, namely the dependence of later states on the preceding states; this dependence differs from purely logical and mathematical relations of dependence in that it is at once a time-relation and a rational relation, since the following state both follows after and follows from the preceding one. Even if the continuity be exhibited by the help of ever so many intermediate members and nuances, this relation continues to hold for every little transition between two members or nuances.

And by this nothing is altered, even if equivalence is exhibited between the states that succeed one another. Hume's problem has by no means, as it has sometimes been said, been solved by the discovery of the conservation of energy. A relation of equivalence does not exclude a qualitative difference, but precisely presupposes it: I have no ground for setting up the judgment $A = B$ when A and B have not first stood before me as different, before I found on closer investigation that they could be put in the place of one

[45] Bosanquet accordingly concedes this too: *At any given moment* we have no choice but to say that the future is conditioned by the past, … effect by cause. *Logic* I. p. 272.

another. When such a relation of identity has been found, it is indifferent whether we go from *A* to *B* or from *B* to *A*; but at the time when it was to be found, we began with *A* or *B* and came by way of investigation over to the other concept. In a judgment we must therefore distinguish between the psychological process by which the judgment comes to be, and in which a determinate difference asserts itself between the initial representation (subject) and the concluding representation (predicate), and the finished judgment, which can be formulated as a relation of identity in which the difference between subject and predicate becomes indifferent.[46] And just as little as in the movement of thought is the order of the members indifferent in outer happening. If I have recognized that there is a relation of equivalence between heat and motion, it is, purely abstractly regarded, indifferent whether I go from heat to motion or from motion to heat; but "the direction of the changes" is in reality a question of life and death: existence actually becomes another thing according as the changes proceed preponderantly in the one or in the other direction. Time then cannot in reality be reversed, and therefore the exhibition of relations of equivalence cannot form the conclusion of our knowledge. We must in addition have information about the actual direction of the changes (the conversions), and that is won only by ever new experience.

There comes in addition that every relation of equivalence — as well as every causal relation whatever — comes into force only when certain conditions, and particularly releasing conditions, are present. With respect to the presence of these conditions a new problem then arises: can it be shown, with respect to them too, that the relation to their cause may be presented as a relation of equivalence? We come here into an infinite series, where the answers (the determinate, concrete answers) cease long before the questions cease. Here too we have a purely logical analogy; for every absolute relation of identity ($A = B$) between two concepts rests, on closer consideration, on determinate conditions (so that it — according to Jevons's formulation — must be expressed by $AC = BC$);[47] it then becomes a new question how matters stand with these conditions (C), and so thought goes ever further. —

There is thus no prospect of escaping the historical element in our knowledge. The measure of the development of knowledge consists in the extent to which the elementary concept of cause (unavoidable succession) can be applied, as compared with the mere ascertaining of actual simultaneous or successive differences, and next in the extent to

[46] See *Det psykologiske Grundlag for logiske Domme.* p. 46–50.

[47] Jevons: *Principles of Science.* 2. ed. London 1877. p. 43. (Already De Morgan, as Jevons remarks, has said that we usually think and argue within a limited world or sphere of concepts, even when this is not expressly declared.) — Compare my *Formel Logik* 3, § 16 and § 22.

which the elementary concept of cause can be replaced by the ideal concept of cause (equivalence or identity). But the process of knowledge will at every time consist in an ascent through the three stages here indicated — a process of ascent which must ever anew be repeated from new starting-points. And it is the reality of the time-relation that conditions this necessity. Speculative conceptions — whether they appear in idealistic or in realistic form — have therefore always an inclination to push the time-relation aside or to regard it as merely "empirical", if not as illusory.[48] Should it be an illusion, then it is in any case an illusion in the second power when one believes one can free oneself from this illusion. For us existence never dissolves wholly into thought.

[48]Francis Bradley has clearly seen the contradiction between time and "the Absolute" when he declares: If time is not unreal, I admit that our Absolute is a delusion (App. and Reality: p. 206). (The Absolute has no seasons. ib. p. 500). — The critique of, and the reaction against, a speculative tendency therefore also often supports itself on the reality of time as a main argument. Compare C. H. Weisse's and S. Kierkegaard's relation to Hegel (*Den nyere Filosofis Historie.* II. p. 243; 261 f.) In his acute work *Tidsexistensens Apologi. Et stycke relationsteori* (Upsala 1888) Pontus Wikner has attempted to shake the either-or that here presents itself. He wants to show that the highest perfection can be attained only by a successive unfolding of qualities which as simultaneous would exclude one another. But the limitation that lies in succession itself is not thereby abolished!

III. The Problem of Existence

IV. The Problem of Valuation

A. The Ethical Problem

a. Ethical Work

b. The Rationality of Ethical Valuation

B. The Religious Problem